

for her research on noncommutative algebras and the general theory of ideals in rings. She emigrated to the US to escape the Nazis.

Ockham, William of, German philosopher (c.1285–c.1349) See p.165

Ørsted, Hans Christian (1777–1851) Danish chemist and physicist who showed that electricity and magnetism are related by observing the needle of a magnetic compass move when close to a wire carrying an electric current. The unit of magnetic induction was named after Ørsted.

Ohm, Georg Simon (1789–1854) German physicist and discoverer of Ohm’s law, which uses the concept of resistance to formulate the relationship of current and voltage. Ohm’s law was so badly received at the time that he resigned as professor. Its value was recognized later.

Olbers, Heinrich Wilhelm (1758–1840) German astronomer and physician who carried out theoretical work on comets, and discovered two asteroids and five comets. Olbers’ paradox, which asks why the sky is dark at night, remained unanswered during his lifetime.

Oppenheimer, Robert (1904–67) American physicist best known as the father of the atomic bomb. Oppenheimer investigated subatomic particles, before becoming director of the Manhattan Project in 1941 under General Groves. Although he won the Presidential Medal of Merit in 1946, Oppenheimer was accused of communism in 1953. He received the Enrico Fermi Award in 1963 as a gesture of reconciliation.

Otto, Nikolaus August (1832–91) German engineer who invented the four-stroke internal combustion engine. Otto’s prize-winning engines offered a practical alternative to steam power. Described theoretically by the Otto cycle, four-stroke engines were used by Karl Benz and Gottlieb Daimler in the first motorcars.

Oughtred, William (1574–1660) English mathematician and teacher who invented the slide rule. His popular and influential textbook *The Key to Mathematics* (1631), introduced symbols such as “x” for multiplication.

Owen, Richard (1804–92) English anatomist and paleontologist who coined the word Dinosauria or “terrible reptile.” Owen published several texts on dinosaurs, classifying them differently from other reptiles. Owen helped establish London’s Museum of Natural History, and although he believed in evolution, was an outspoken opponent of Darwin’s theory.

Papin, Denis (1647–c.1712) French physicist and inventor whose steam digester led to the development of steam engines. Papin also invented a steam safety valve, a condensing pump, and a paddle-wheel boat.

Paracelsus (1493–1541) Swiss physician and alchemist who established the use of chemistry in medicine. Traveling and practicing medicine across Europe, Paracelsus introduced laudanum, sulfur, lead, and mercury as medicinal remedies and gave a clinical description of syphilis. An outspoken opponent of university medicine, he gained huge influence by writing and speaking in German.

Pascal, Blaise, French mathematician and physicist (1623–62) See p.107

Pasteur, Louis, French chemist, biologist, and microbiologist (1822–95) See p.214

Pauli, Wolfgang (1900–58) Austrian-born American theoretical physicist awarded the 1945 Nobel Prize in Physics for his Pauli exclusion principle, which states that no two electrons in an atom can exist in the same quantum state simultaneously. Pauli also devised an atomic model of the thermal properties of metal, and was the first to propose the existence of neutrinos.

Pauling, Linus Carl, American chemist (1901–94) See p.271

Pavlov, Ivan Petrovich (1849–1936) Russian physiologist whose experiments on dogs led to his discovery of the conditioned reflex. Pavlov showed that dogs salivate in anticipation of food, not just at the sight of it. He was awarded the 1904 Nobel Prize in Physiology or Medicine and summarized his work on behaviorism in the 1926 book, *Lectures on Conditioned Reflexes*.

Perkin, William (1838–1907) English chemist celebrated for creating the first synthetic dye, aniline purple, which became extremely fashionable. While synthesizing quinine, Perkin came across a bluish dye now called aniline purple, which he patented and manufactured, enabling him to retire at 35.

Petit, Alexis Therese (1791–1820) French physicist who discovered the Dulong–Petit Law with Pierre Dulong. This states that for all solid elements, the specific heat multiplied by the atomic weight is a constant. He also designed a thermometer to measure the dilation coefficients of metals.

Planck, Max, German physicist (1858–1947) See p.236

Plato, Greek philosopher (424 BCE–348 BCE) See p.25

Poincaré, Henri, French mathematician (1854–1912) See p.227

Priestley, Joseph (1733–1804) English chemist and clergyman who discovered several gases including the one later identified as oxygen. Learning the theories of electricity from Benjamin Franklin, Priestley began his own electrical experiments and presented his findings in the popular 1767 work *The History and Present State of Electricity*. He then experimented with gases and made important discoveries, although he believed in the phlogiston theory that was later discarded.

Proust, Joseph-Louis (1754–1826) French chemist best known for formulating the law of definite proportions (Proust’s law), which states that in any compound, the elements are present in a fixed proportion by weight.

Ptolemy (Claudius Ptolemaeus) (c.100–c.170 CE) Greek astronomer and geographer whose Ptolemaic system placed Earth at the center of the Universe and incorporated complex epicycles. Based in Alexandria, Ptolemy also made a map of the world and wrote the encyclopedia *Almagest*.

Pythagoras (580–500 BCE) Greek philosopher and mathematician whose teachings contributed to mathematics and rational philosophy. Pythagoras taught that nature and the world could be interpreted through

numbers, and greatly influenced Plato and Aristotle. He is credited with discovering the chief musical intervals and the Pythagorean theorem of geometry.

Raman, Chandrasekhara Venkata (1888–1970) Indian physicist awarded the 1930 Nobel Prize in Physics for work on the scattering of light, called the Raman effect. This shows that when light passes through transparent material, a small proportion of the deflected light changes in wavelength (that is, in energy).

Ramón y Cajal, Santiago, Spanish histologist and neuroscientist (1852–1934) See p.229

Ramsay, William (1852–1916) Scottish chemist awarded the 1904 Nobel Prize in Chemistry for discovering the inert gases argon, neon, xenon, and krypton. He also discovered the rare gas radon and isolated helium from liquid air.

Ray, John (1627–1705) English naturalist and botanist whose contributions helped found modern taxonomy. A fellow of Cambridge’s Trinity College, Ray lost his position during the Restoration and began studying botany and zoology across Europe. He set out his classification of plants in *Historia Plantarum*, which established the species as the basic unit of taxonomy.

Réaumur, René, French physicist and entomologist (1683–1757) See p.134

Rhazes (see Al-Razi)

Richter, Charles (1900–85) American physicist, seismologist, and developer of the logarithmic Richter scale, which records the magnitude of an earthquake at its epicenter. Richter also devised a map of the most earthquake-prone areas in America.

Rømer, Ole Christensen (1644–1710) Danish astronomer who established that light travels at a finite speed. Rømer calculated the speed of light to be 140,000 miles (225,000 km) per second, around 47,000 miles (75,000 km) per second slower than modern estimates. Rømer also invented a temperature scale and introduced the first Danish system for weights and measures.

Röntgen, Wilhelm (1845–1923) German physicist and recipient of the first Nobel Prize in Physics in 1901 for discovering X-rays. A professor of physics, Röntgen researched elasticity, capillarity, polarized light, and the specific heat of gases. His 1895 discovery of X-rays was enormously important for medicine and modern physics.

Rumford, Benjamin Thompson (1753–1814) American-born British physicist, inventor, soldier, and administrator, best known for his work on heat. Rumford correctly theorized that heat was produced by the motion of particles, rather than being a liquid form of matter as thought previously. He helped to found the Royal Institution of London in 1799.

Russell, Henry Norris (1877–1957) American astronomer who helped establish the modern science of theoretical astrophysics. Russell is known for discovering the relationship between a star’s magnitude and its spectral type, which he presented in the 1910 Hertzsprung–Russell diagram. He also theorized an abundance of hydrogen in stellar

atmospheres—now considered a fundamental tenet of modern cosmology.

Rutherford, Ernest, New Zealand-born chemist and physicist (1871–1937) See p.248

Salam, Abdus (1926–96) Pakistani nuclear physicist and co-winner of the 1979 Nobel Prize in Physics for formulating the electroweak theory, which unifies the weak nuclear force and electromagnetic interactions of elementary particles. Salam was a professor of theoretical physics in London, and the first Muslim scientist to win a Nobel prize.

Salk, Jonas Edward (1914–95) American physician and medical researcher who discovered the first effective vaccine for polio. After working on an influenza vaccine at Michigan University, Salk began human trials of his polio vaccine in 1952. In 1955, the vaccine was released for use in America, virtually eradicating polio.

Sanger, Frederick (1918–) English biochemist and the only person to be awarded the Nobel Prize in Chemistry twice. Sanger’s 1958 Prize was for his research on the structure of proteins, in particular insulin. His 1980 Prize was for his method of sequencing DNA molecules, which was used to develop the first fully sequenced DNA-based genome.

Schrödinger, Erwin (1887–1961) Austrian theoretical physicist and co-recipient of the 1933 Nobel Prize in Physics for his contributions to quantum mechanics. Schrödinger is best known for his equations of wave mechanics, but his book *What is Life?* (1948) greatly influenced molecular biology.

Schwann, Theodor (1810–82) German physiologist who founded histology by proposing that all organisms are composed of cells. He discovered the digestive enzyme pepsin and the cells that surround nerve axons. He helped disprove the theory of spontaneous generation, and coined the term “metabolism.”

Semmelweis, Ignaz (1818–65) Hungarian physician who pioneered the use of antisepsis to prevent deaths caused by puerperal fever. Although he showed that the high mortality rates associated with childbirth could be reduced by physicians washing their hands in chlorinated lime, this practice was not introduced until many years later.

Servetus, Michael, Spanish physician (c.1511–53) See p.80

Shen Kuo (1031–95) Polymathic Chinese scholar who discovered magnetic declination and described the first magnetic needle compass. Shen’s finding is one of the many recorded in his famous book, *Brush Talks from Dream Brook*. He also described movable type, formulated a geological hypothesis about fossils, and undertook an ambitious project of mapping the stars.

Shockley, William Bradford (1910–89) American physicist who shared the 1956 Nobel Prize in Physics with John Bardeen and Walter Brattain for inventing the transistor. Professor of Engineering at Stanford University, Shockley commercialized his transistor, which led to the development of California’s Silicon Valley. He later caused controversy by advocating eugenics and proposing sterilization for those with low IQs.